Presentation and challenge session

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Presentation and challenge session before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Final project manuscript

Without opening the old lesson, explain one key idea from Final project manuscript. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Peer review and response

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Figures that preserve evidence

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Presentation and challenge session trace and record the first step you can explain without help.

Use W2: Trace the worked case for Presentation and **challenge session**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Presentation and challenge session explanation against the supplied facts before discussing it.

Use the worked case for Presentation and **challenge session**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Presentation and challenge session facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Presentation and challenge session facts and keep your reasoning.

A presenter shows one perfect **output** after a20-case study with14 passes. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Presentation and challenge session facts and keep your reasoning.

A report omits this qualification: A confident response should not replace missing **evidence** with speculation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Presentation and challenge session facts and keep your reasoning.

Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Presentation and challenge session facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Presentation and challenge session to Reproducible AI Research Study, identify a limit and choose the next useful test.

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Presentation and challenge session

> Answer a challenge without inventing evidence. At the presentation, a reviewer asks whether the model works on a different camera. Explain the labelled step before checking the changed condition.

Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.

The project tested twelve photos from camera A only. Ten were correct; no camera B records exist.

AI ACADEMY / AI-7/35

Presentation and challenge session

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. State measured result

Report 10/12 correct on camera A.

2. Address the challenge

Say camera B performance is not measured.

3. Propose a valid next test

Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

CHANGE ONE THING

The reviewer shows one camera-B success.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

1. Propose a valid next test

2. State measured result

3. Address the challenge

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Propose a valid next test Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The reviewer shows one camera-B success. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. That is one demonstration, not an estimate of broad camera-B reliability.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That is one demonstration, not an estimate of broad camera-B reliability. One selected **observation** does not represent the full proposed condition.

Repair a missing building block

Review Final project manuscript (ai-7/34). Goal: Explain the **mechanism** and terminology of final project manuscript. Short practice: Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping.

Which abstract claim must be corrected if it says ten independent plants?. Return to this week and explain what you changed.

Review Peer review and response (ai-7/32). Goal: Explain the **mechanism** and terminology of peer review and response. Short practice: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?. Return to this week and explain what you changed.

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

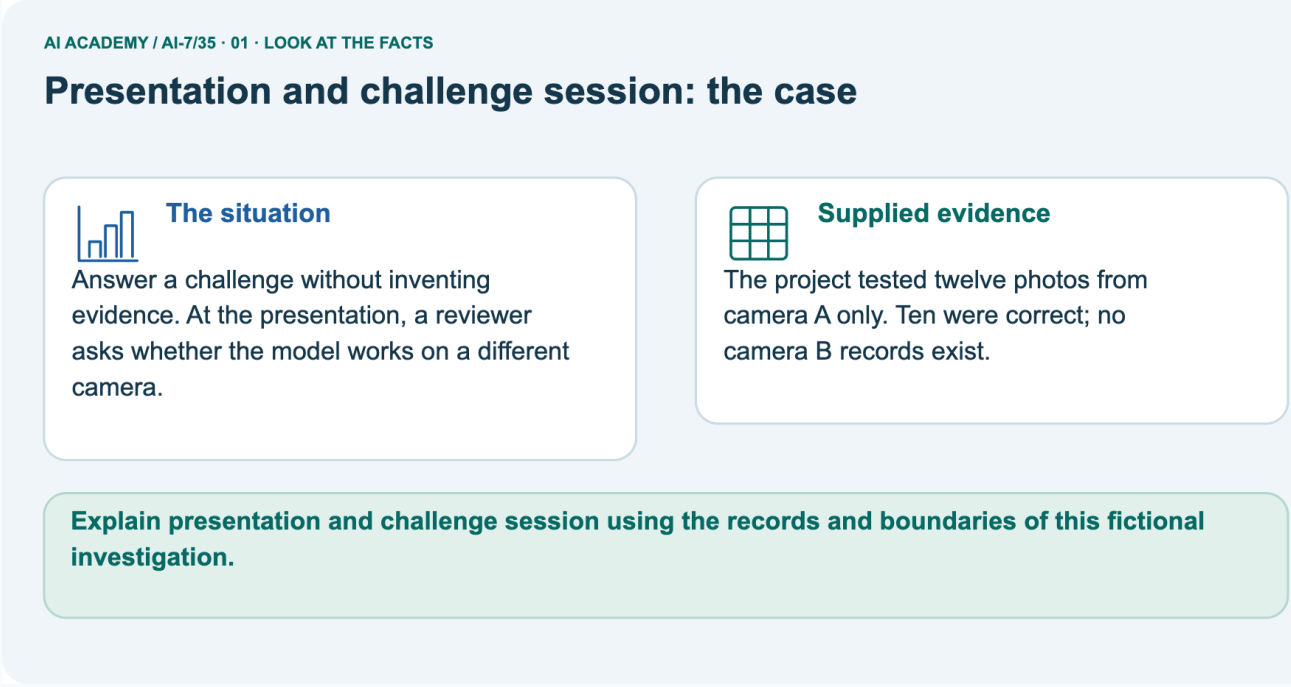
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

Explain each Presentation and challenge session picture from its labels, then record a relevant limit.

01. Presentation and challenge session: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

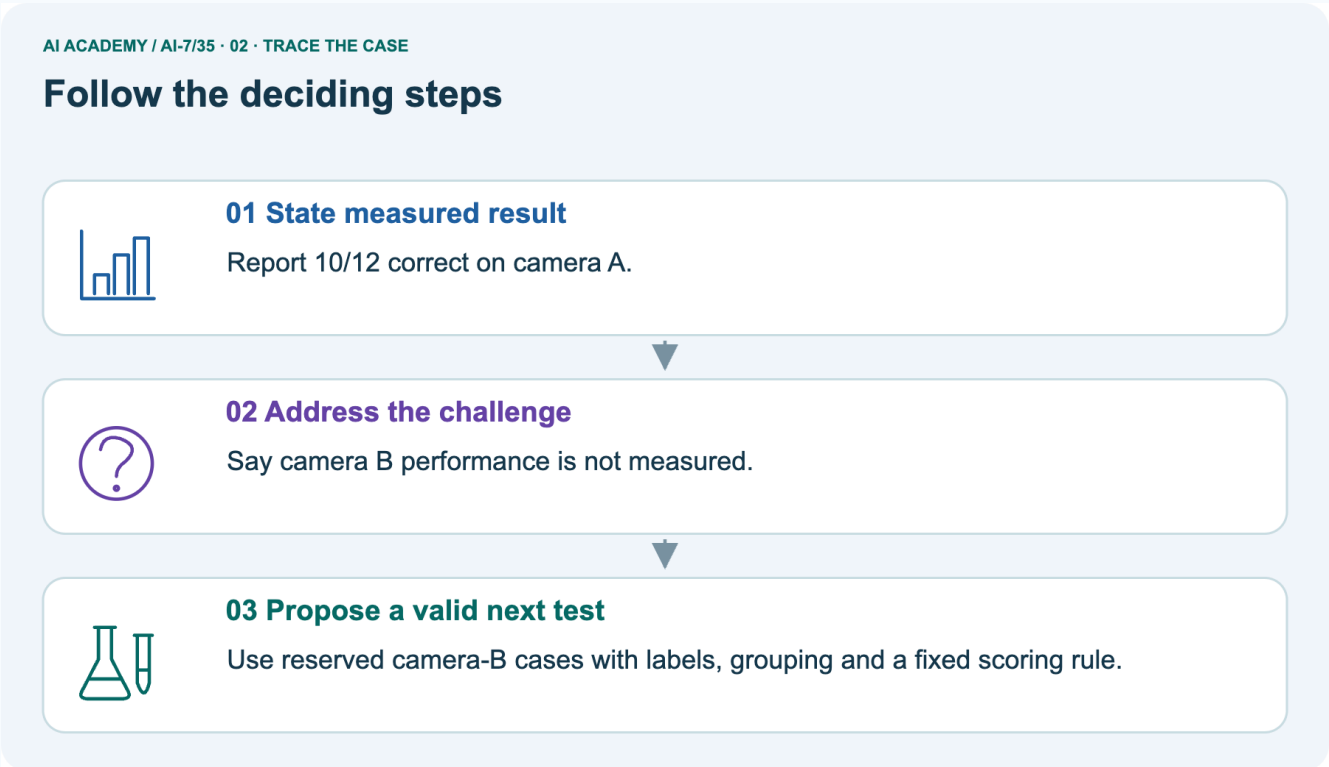
Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera. Supplied facts: The project tested twelve photos from camera A only.

Ten were correct; no camera B records exist.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



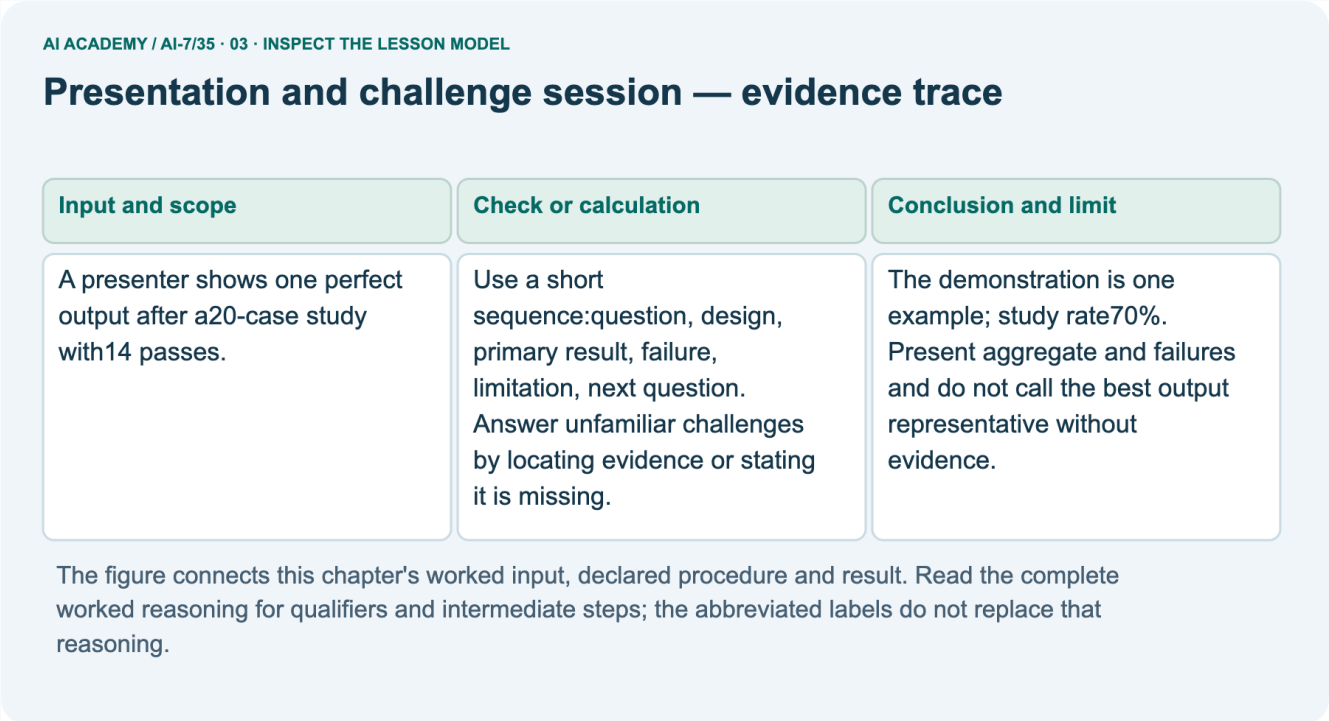
Follow the labelled picture with the matching explanation. Open the vector diagram

State measured result: Report 10/12 correct on camera A. → Address the challenge: Say camera B performance is not measured. → Propose a valid next test: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Presentation and challenge session — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

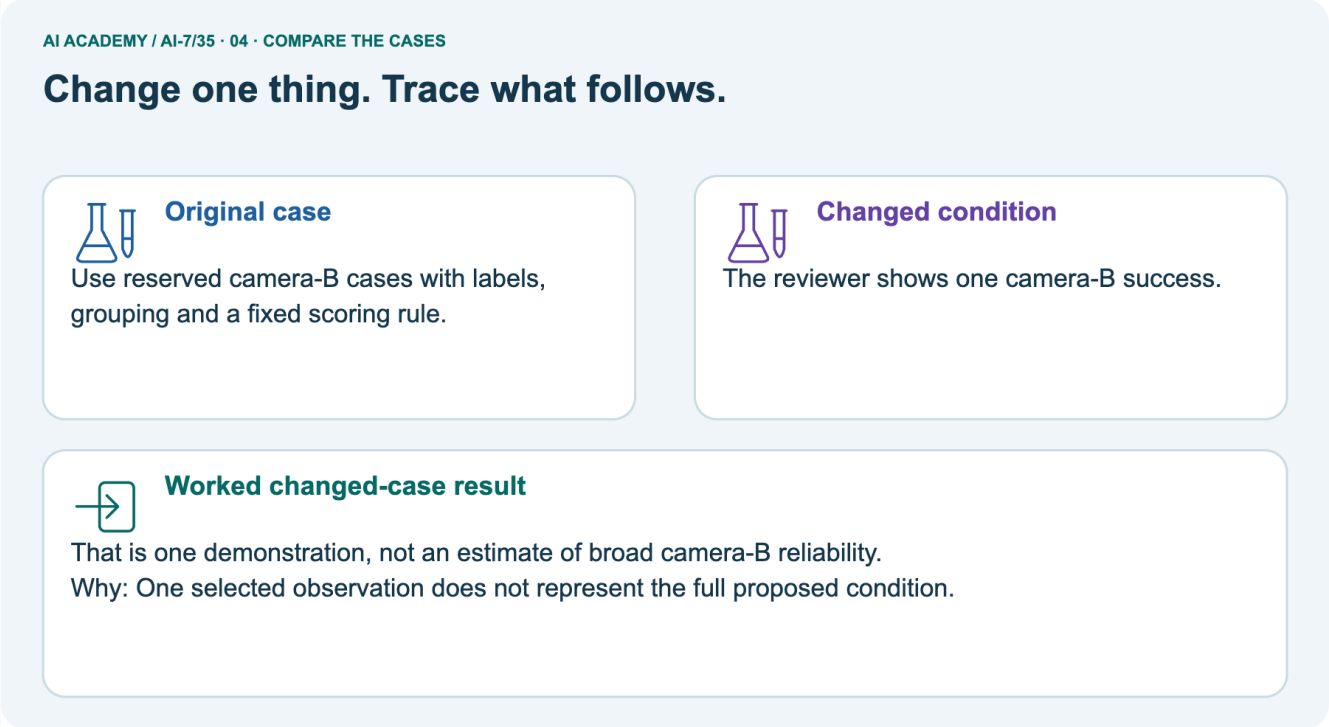
Presentation and **challenge session** — **evidence** trace. A presenter shows one perfect **output** after a20-case study with14 passes. | Use a short sequence:question, design, primary result, failure, limitation, next question.

Answer unfamiliar challenges by locating **evidence** or stating it is missing. | The demonstration is one example; study rate**70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

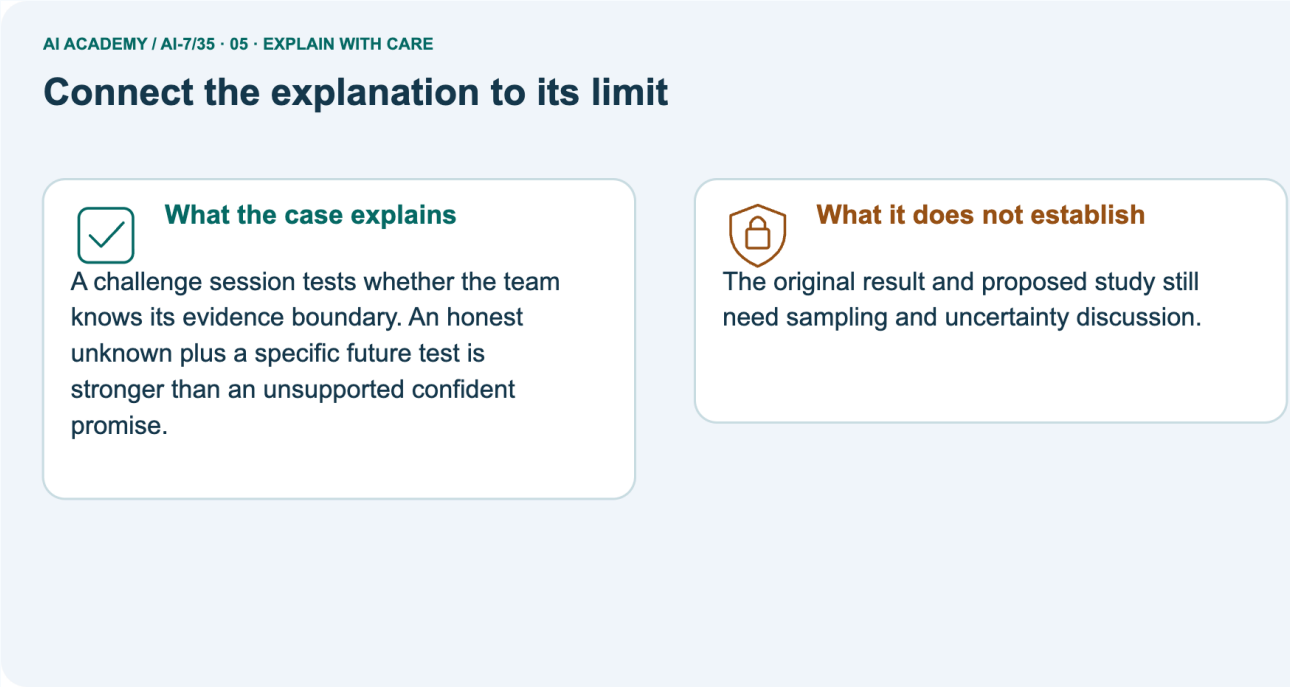
Original case: Use reserved camera-B cases with labels, grouping and a fixed scoring rule. Changed condition: The reviewer shows one camera-B success.

Expected result: That is one demonstration, not an estimate of broad camera-B reliability. Reason: One selected **observation** does not represent the full proposed condition.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

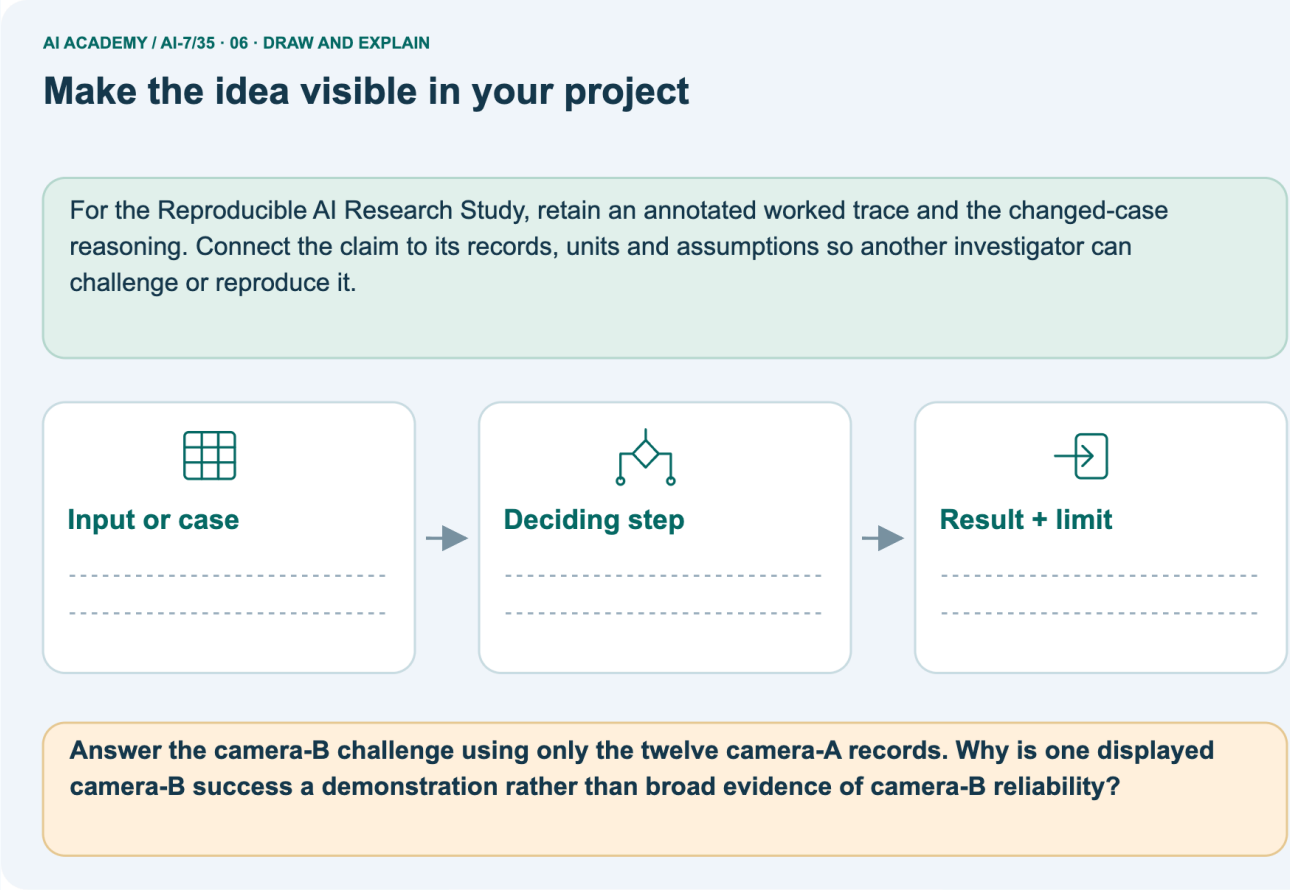
A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

Limit: The original result and proposed study still need sampling and uncertainty discussion.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Answer the camera-B challenge using only the twelve camera-A records.

Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.